Straddles and strangles on the VRP-band book: more P&L, not less volatility

Experiment report · L/S Gamma Desk, QUANTT

2026-11-13

Question: on the VRP-band rules, does trading an ATM straddle for long gamma and a 20-delta strangle for short gamma (1 contract per leg) make the book's daily net P&L less volatile than the single ATM call or put it trades now, without a deeper drawdown, a worse tail, or more than $1,000 less net P&L?

Method:

- Pre-registered backtest: the hypothesis card and pass bar were committed before any run (commit 7d70155).
- The VRP-band research baseline (`configs/paper.toml` + `configs/backtest.toml`) is replayed on R2 replay snapshots v1 (config_hash b040760187f0) over the development window, 2014-06-02 to 2024-12-31. The out-of-sample window (2025-01-01 to 2026-07-10) was not used.
- Variants:
  - `both` (primary): straddle on long signals, 20-delta strangle on short signals.
  - `straddle_only`: straddle on long signals; short signals keep the ATM put.
  - `strangle_only`: strangle on short signals; long signals keep the ATM call.

  Comparison: the registered baseline run matching the current config, at the same spread multiple.
- Every variant was run at 1x and 2x spreads. Costs:
  - $0.65 per contract and $0.005 per share, with a $1 minimum per order;
  - a structure's two legs are charged as one order;
  - each leg fills at its own quoted half-spread.

  Signals, exits and hedges run once a day at 10:15 ET.
- A structure exits as one position: the `paper.toml` exit rules are applied to the legs' combined P&L (options plus hedge shares) and combined premium, and both legs are sent to close together. One exception is in the caveats.

1 Result vs the pass bar

Every check had to hold at 1x and 2x. "Vol" is the standard deviation of daily net P&L in USD, flat days included. ES is the mean of the worst 5% of daily net P&L. Everything is at one contract per leg.

Check (`both` vs baseline)	1x value	1x bar	2x value	2x bar	Result
Vol, full window, lower	$54.37	$35.64	$54.75	$37.28	fail
Vol, 2014-2019, lower	$25.80	$16.31	$26.51	$17.84	fail
Vol, 2020-2024, lower	$74.30	$48.94	$74.61	$50.88	fail
Max drawdown, lower	$3,848	$3,027	$4,405	$4,258	fail
ES 5% magnitude within 1.10x	−$137.8	−$106.1	−$142.3	−$115.0	fail
Net P&L ≥ baseline − $1,000	−$318	−$3,060	−$1,855	−$5,122	pass
Event floor: 20 straddles and 20 strangles (1x)	30 and 102				met

Verdict: failed. The two-leg book is more volatile than the single-leg book in every period and at both spread levels, and its drawdown and tail are worse. It lost less money: −$318 against −$2,060 at 1x, and all of that gain came from the strangle. No variant made money at 2x except `strangle_only` (+$302).

1.1 Full-window numbers (not part of the bar)

	baseline 1x	both 1x	straddle_only 1x	strangle_only 1x	both 2x	strangle_only 2x
Net P&L	−$2,060	−$318	−$3,783	+$1,398	−$1,855	+$302
Vol (daily std)	$35.64	$54.37	$48.40	$43.54	$54.75	$43.96
Sharpe	−0.36	−0.04	−0.48	0.20	−0.21	0.04
Max drawdown	$3,027	$3,848	$4,765	$2,435	$4,405	$2,754
ES 5% (daily)	−$96.5	−$137.8	−$122.6	−$114.7	−$142.3	−$118.0
Structures (long / short)	30 / 101	30 / 102	30 / 101	30 / 102	30 / 102	30 / 102
Commissions (options / hedges)	$262 / $914	$343 / $878	$280 / $916	$325 / $876	$343 / $878	$325 / $876

The registry's `trades` column counts legs (264 for `both`), so the table counts structures instead.

2 Charts

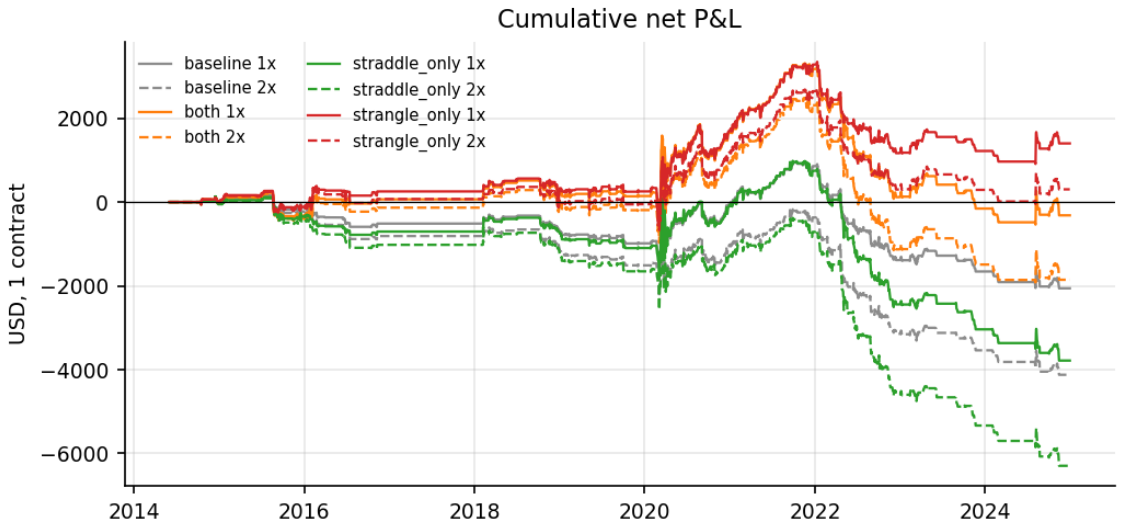


Figure 1: Cumulative net P&L, 1x solid and 2x dashed; baseline grey

The strangle variants pull ahead in 2020 and 2021 and give back part of it in 2022. `straddle_only` tracks below the baseline from 2022 on.

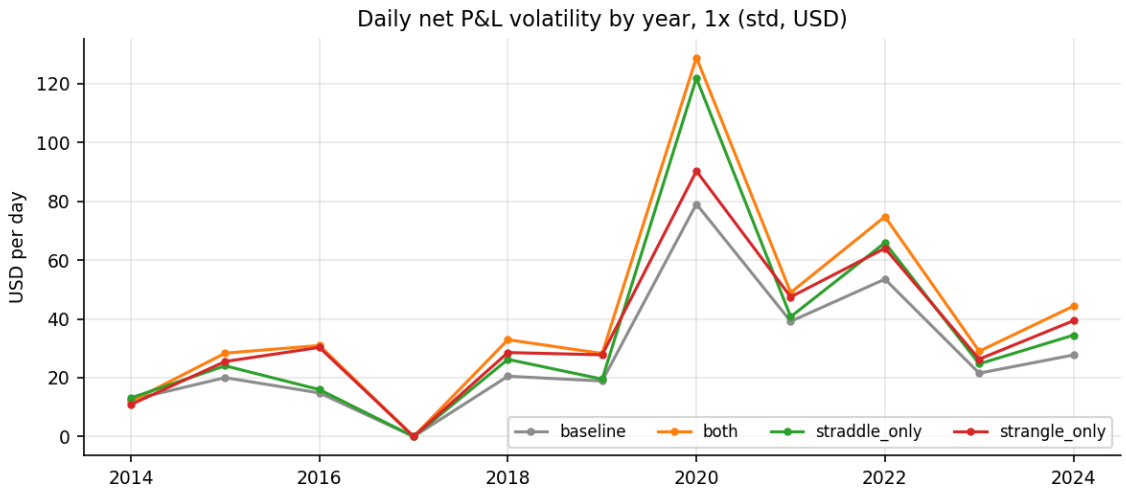


Figure 2: Daily net P&L volatility by year, 1x

`both` is above the baseline in every full year that traded (2017 had no positions; 2014 is a half year, $11.9 against $12.5), and most of all in 2020. Each half adds volatility on its own.

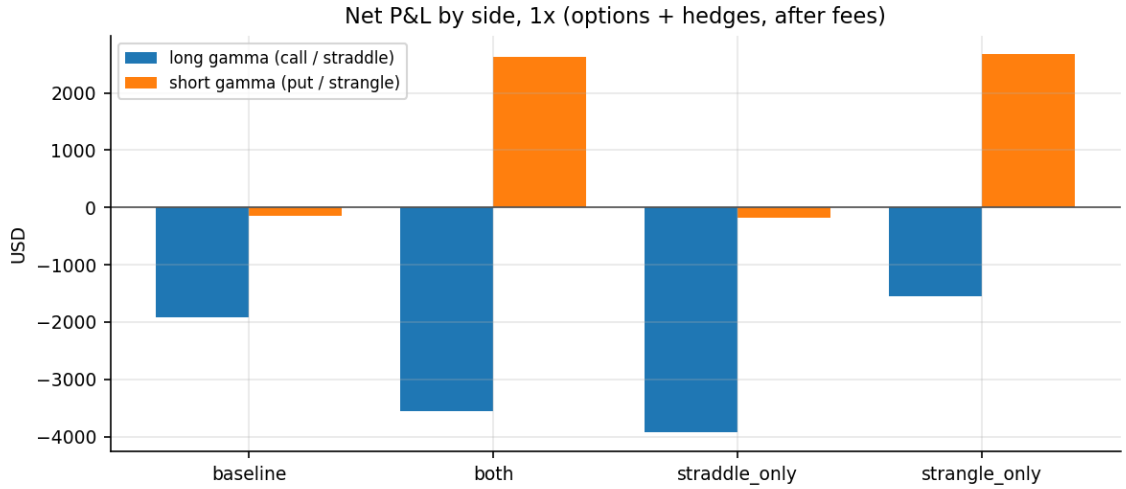


Figure 3: Net P&L by side, 1x

The long side loses more with a straddle (−$3,555 in `both` against −$1,913 for the call). The short side turns from −$148 with the ATM put to +$2,632 with the strangle.

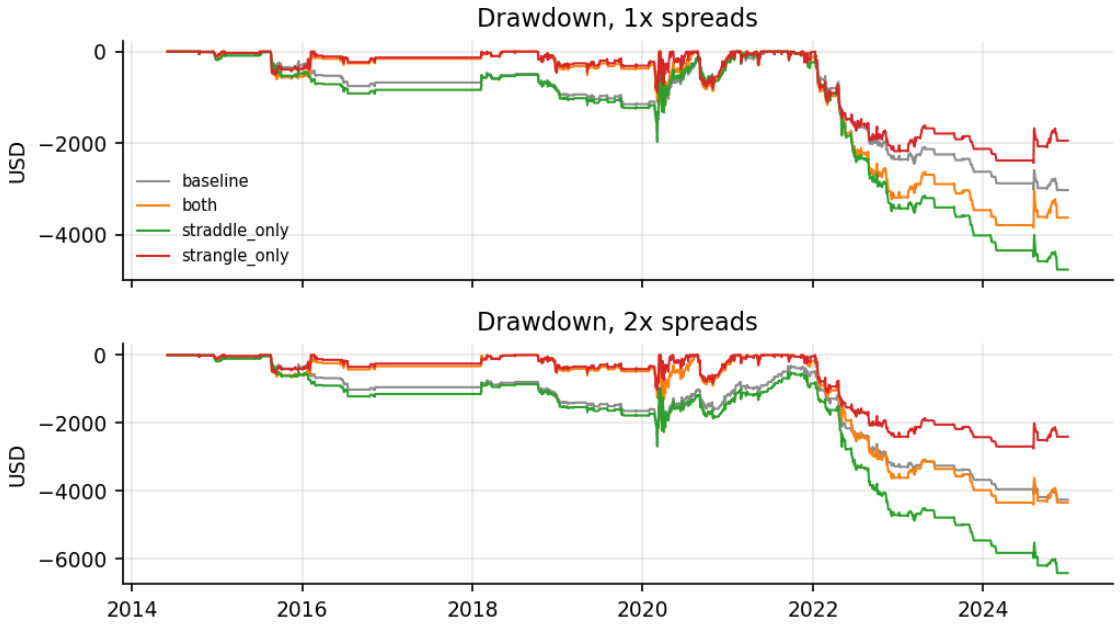


Figure 4: Drawdown at 1x and 2x spreads

Most of the drawdown in every variant is 2022 and after. `strangle_only` has the shallowest drawdown at both multiples, and `straddle_only` the deepest.

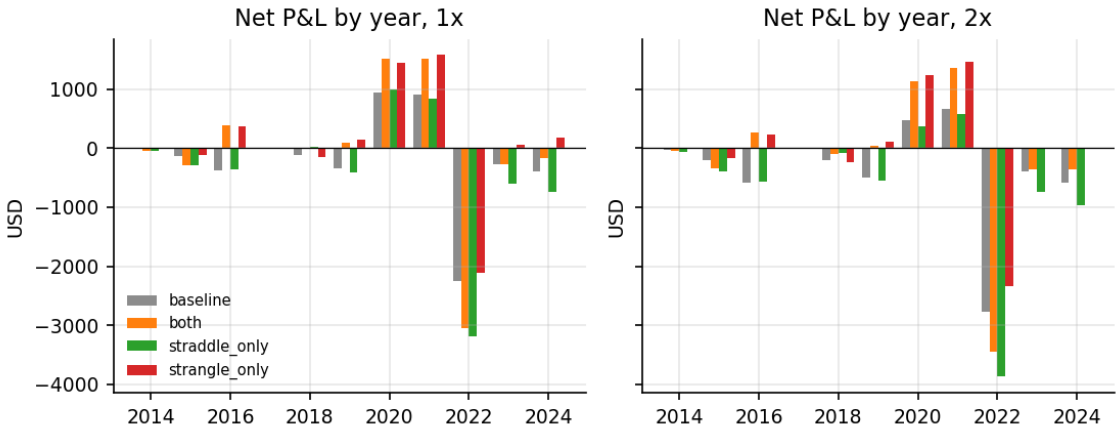


Figure 5: Net P&L by year

2022 is the worst year for every variant. The straddle makes 2022 worse (`both` −$3,056 against −$2,242 for the baseline at 1x, by day).

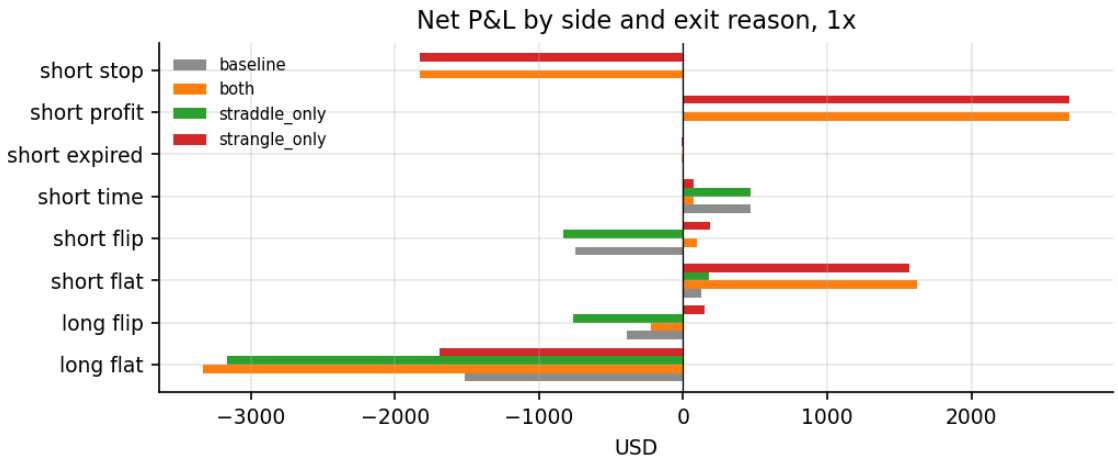


Figure 6: Net P&L by side and exit reason, 1x

Bars are per leg, by each leg's own exit reason. Counted per structure, strangles hit the 50% profit take 9 times (+$2,670) and the 1x-credit stop 4 times (−$1,827). The ATM put hit neither in 101 round trips; it closed on flat, flip and time.

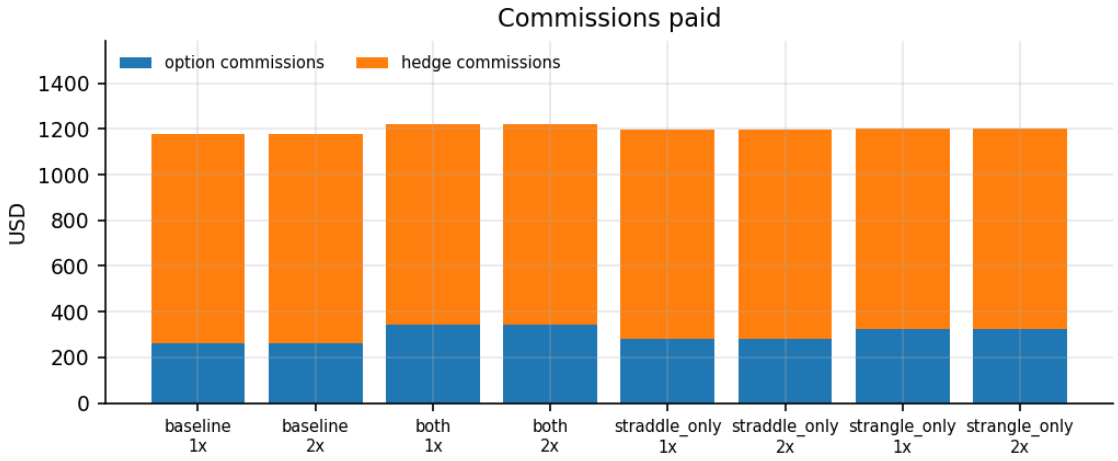


Figure 7: Commissions paid

The second leg adds $81 of option commission over the window in `both` at 1x. Hedge commission falls slightly with the strangle ($876 against $914), which opens close to delta-neutral. `straddle_only` pays about the same ($916).

3 Analysis

The volatility result is mechanical. At one contract per leg, both structures carry more gamma and vega than the single leg they replace. This was measured from the entry snapshot for every structure at 1x:

- A straddle averages 2.0x the gamma and 2.0x the vega of the ATM call. It also costs about 2x the premium.
- A 20-delta strangle averages 1.4x the gamma and 1.4x the vega of the ATM put, for about 60% of the put's credit.

More vega means larger daily P&L swings, whatever the delta hedge does. Vol rose in every full year that traded (2015-2024, except 2017 with no positions) for `straddle_only` and `strangle_only` alike, so neither half delivers the hypothesis. The worst day in all four books is 2020-03-20. The straddle made that day worse (−$786 against −$468); the strangle did not change it.

The P&L gain is the strangle's exit rules, not its lower gamma. The baseline's ATM put never reached the 50% profit take or the 1x-credit stop in 101 round trips. A strangle's credit is smaller, and both legs decay while SPY stays between the strikes, so the 50% target was reached 9 times. Those 9 exits made +$2,670, more than the short side's whole net P&L (+$2,632). The 4 stops cost −$1,827.

The result is concentrated:

- The 5 best strangles made $1,933 of the short side's $2,632.
- Without them, the short side is +$699.
- By year, most of the gain is 2020 and 2021 (+$2,324 of the $2,632, by exit date).

The straddle loses more for the same reason the call does. Long signals fire when IV is below the RV forecast. The 2026-10-16 short-only card found that the long calls lose money on this window anyway. A straddle doubles that bet's size: 30 straddles lost −$3,555 against −$1,913 for 30 calls. The loss is concentrated in straddles closed in 2022 (−$2,011) and 2023-2024.

Path dependence:

- All 30 long entries fall on the same dates in every variant.
- Short entries differ: 93 of the baseline's 101 short entry dates are shared with `both`, because profit takes and stops end positions earlier and the next entry moves.
- The long side of `strangle_only` (−$1,545) differs from the baseline's (−$1,913) with the same calls, because exits and re-entries around the strangles change when positions end.

Differences of a few hundred dollars between variants are within that noise.

4 Verdict

Failed, per the pre-registered bar: volatility, drawdown and the tail are all worse at 1x and 2x, and only the P&L floor held. As specified (one contract per leg), the two-leg structures add risk rather than remove it. Archive this card.

The most interesting side result is `strangle_only` (+$1,398 / +$302). It was reported only and was not judged by this bar. Its edge rests on 9 profit takes and the 2020-2021 years, and it is the only variant positive at 2x. A follow-up would need its own card: for example, a 20-delta strangle against the ATM put on net P&L and Sharpe, with gamma matched, so that size is not the difference.

5 Caveats

- **In-sample:** the idea follows the short-only, trend-long-band and lower-delta results on this same window. Only the out-of-sample run can confirm any of it.
- **Size:** one contract per leg, so the structures are not risk-matched with the baseline. The volatility comparison mostly measures size.
- **Fills:** each leg is filled separately at its own half-spread. A combo order would usually fill better, so the 2x run is the stress test.
- **Spread model:** fitted on about two weeks of calm 2026 live quotes.
- **Data gaps:** R2 has no month-end weekdays, so about 3.3% of scheduled times are skipped.

- **One split structure:** 1 of 102 strangles (early 2016) closed its legs separately. One leg had no mark and could not be filled for about two weeks, so it stayed open alone until expiry; the book also went unhedged on those 14 days. The effect was about −$11.
- **Few events:** 30 straddles, 9 strangle profit takes and 4 strangle stops drive the differences.

6 Trials

6 trials for this experiment (3 variants × 1x and 2x, one run of `experiment.py`, commit 7d70155), plus the baseline's 4 registered runs. The comparison used the 2026-11-11 baseline pair, the first that matches the current config (it reproduces the original baseline exactly). There were no rebuilds and no `-dirty` rows. The pass bar was not changed.